

Robust Inductive Wireless Electric Vehicle Charging

Resonant Design, Loss Budget and Efficiency Analysis

30 kHz series-series compensated inductive power transfer system
MATLAB Simulink / Simscape Electrical model with analytical design study

Document	Design and analysis report (repository companion)
Project	Robust-Inductive-Wireless-Electric-Vehicle-Charging
Author	Rishi P
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Model file	simulation/Resonant_Power_Transfer.slx
Rated reference point	7.956 kW (390 V, 20.4 A) at 30 kHz, $k = 0.327$
Design efficiency target	90 % (analytical loss budget)

This report consolidates the Simulink design, the captured waveforms, and the analytical calculations used to establish the resonant tuning, the coupling coefficient, the equivalent load, the loss budget and the efficiency of the system. Every figure, parameter and equation is taken from the repository files listed in Section 12.

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1. Introduction and scope

Inductive power transfer (IPT) moves energy across an air gap through the magnetic coupling of a transmitter coil and a receiver coil. To transfer useful power at a practical current, both coils are tuned with series capacitors so that each loop is resonant at the inverter switching frequency. This project implements such a system for electric-vehicle charging: a high-frequency full-bridge inverter drives a resonant transmitter coil, the receiver coil feeds a diode bridge, and the rectified output charges a battery model.

The work has two complementary layers. The first is the switching-level Simulink and Simscape Electrical model (*Resonant_Power_Transfer.slx*), whose waveforms are reproduced in Section 10. The second is the set of analytical calculations in this report, which use the parameters of that model to derive the resonant frequencies, the coupling coefficient, the equivalent AC load, the reflected impedance, the conduction-loss budget and the efficiency. Where a number is an analytical estimate or a design target it is labelled as such.

All symbols are defined where first used. Unless stated otherwise, voltages and currents are RMS values of the fundamental component.

2. System architecture

Figures 1 and 2 show the reference architecture of a road-powered charging system. The road side contains the high-frequency inverter, a current-gain and compensation network and the transmitting coil. The vehicle side contains the receiving coil, a voltage-gain and compensation network, a high-frequency rectifier and the battery. Figure 2 contrasts an elongated rail-type transmitter with segmented transmitter coils that are energised in sequence as the vehicle passes.

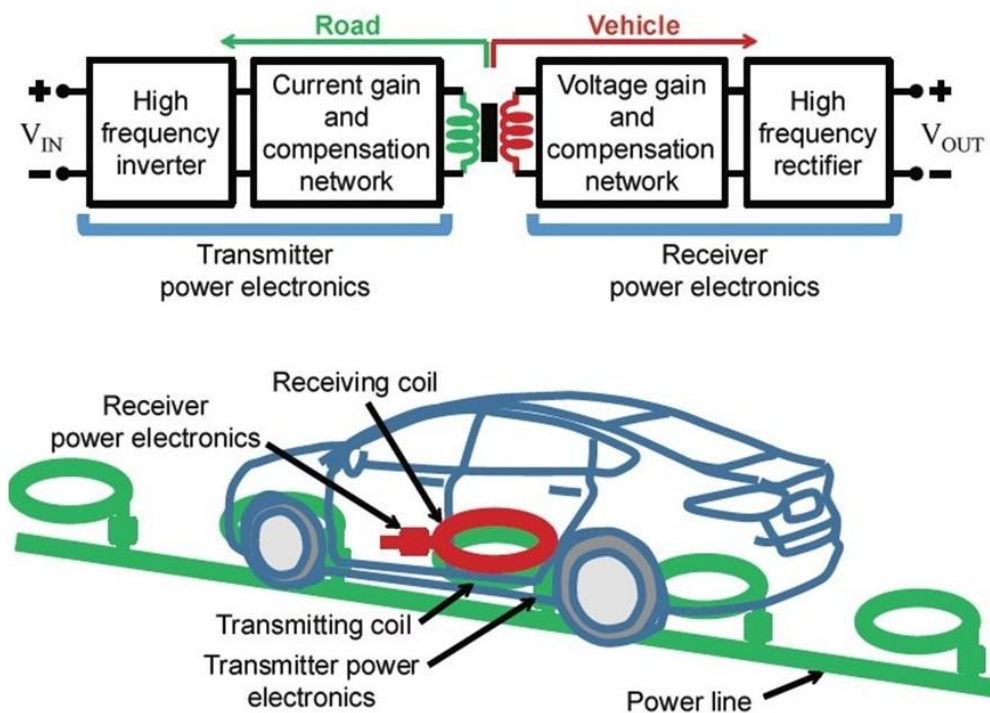


Figure 1. Block diagram of a road-to-vehicle inductive charging system and its physical arrangement (transmitter power electronics, power line, transmitting and receiving coils).

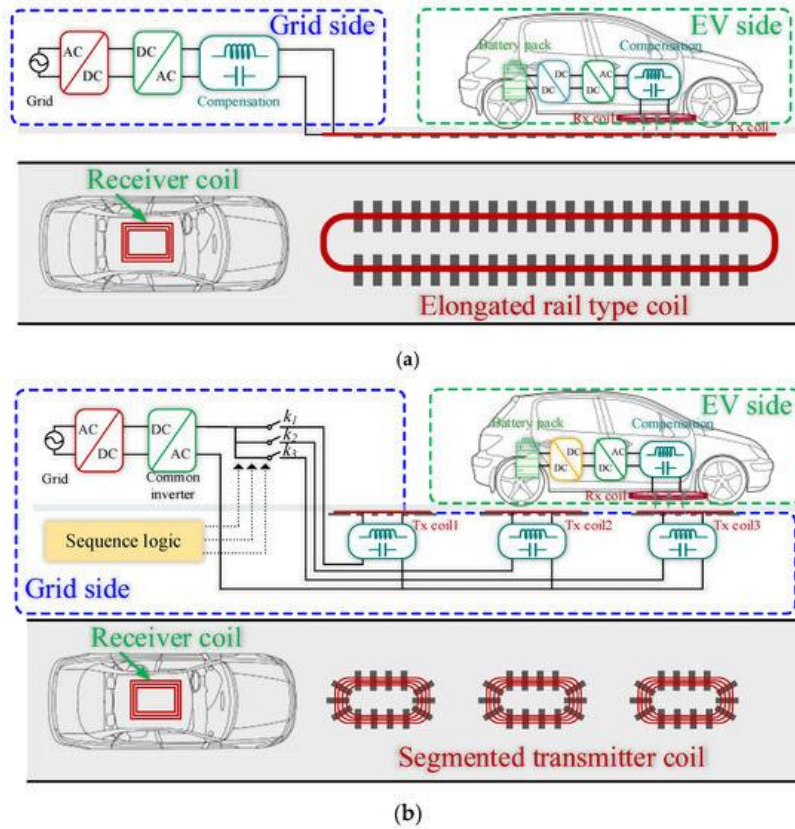


Figure 2. (a) Elongated rail-type transmitter coil. (b) Segmented transmitter coils with common inverter and sequence logic.

The implemented chain follows the same functional structure:

Source (60 Hz, 500 V peak) → diode front-end rectifier → DC link → 30 kHz full-bridge inverter → primary series-resonant network and transmitter coil → magnetic air gap → receiver coil and secondary series-resonant network → diode bridge → battery.

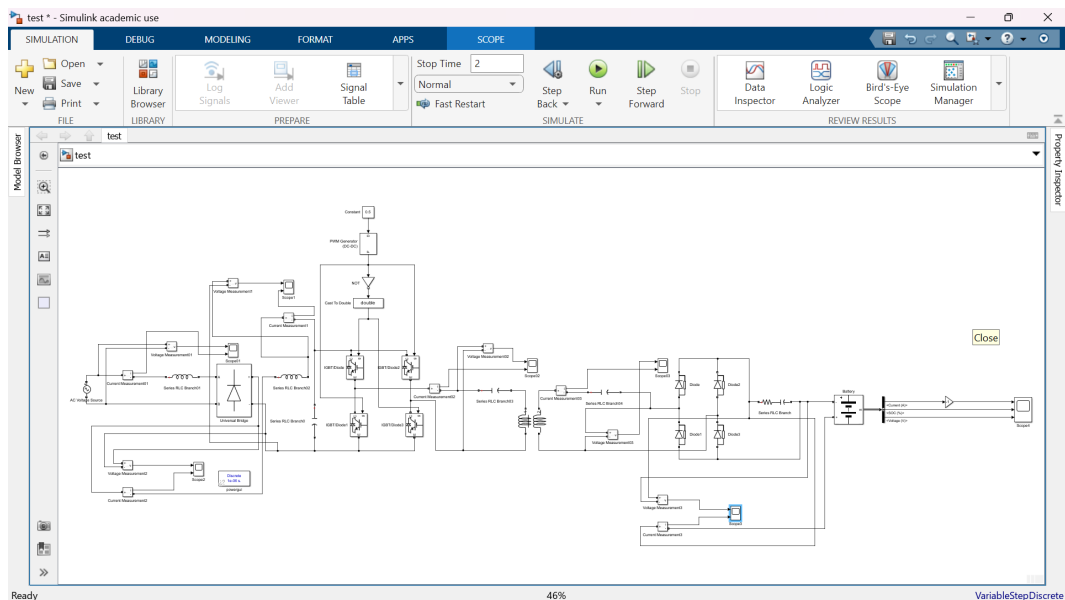


Figure 3. System block diagram of the implemented wireless charging chain (repository figure).

3. Implemented Simulink model parameters

The table lists the parameters read from the saved model file. The coil self-impedances are entered as [resistance, inductance] pairs in the Mutual Inductance block.

Stage	Block	Parameter values
Source	AC Voltage Source	500 V peak (353.55 V rms), 60 Hz
Front-end rectifier	Universal Bridge (diodes), 2 arms	$R_{on} = 1 \text{ m}\Omega$, forward voltage = 0 V
DC-link capacitor	Series RLC Branch0 (C)	300 μF
HF inverter	4 x IGBT/Diode; PWM Generator (DC-DC)	$R_{on} = 1 \text{ m}\Omega$; $F_{sw} = 30 \text{ kHz}$; duty input 0.5
Primary coil	Mutual Inductance, SelfImpedance1	$R_p = 1 \text{ m}\Omega$, $L_p = 266.16 \text{ }\mu\text{H}$
Secondary coil	Mutual Inductance, SelfImpedance2	$R_s = 1 \text{ m}\Omega$, $L_s = 256.79 \text{ }\mu\text{H}$
Mutual inductance	Mutual Inductance, MutuallImpedance	$M = 85.46 \text{ }\mu\text{H}$
Primary capacitor	Series RLC Branch03	$C_p = 105.74 \text{ nF}$
Secondary capacitor	Series RLC Branch04	$C_s = 109.69 \text{ nF}$
Output rectifier	4 x Diode	$R_{on} = 1 \text{ m}\Omega$
Output branch	Series RLC Branch (RC)	$R = 0.1 \text{ }\Omega$, $C = 300 \text{ }\mu\text{F}$
Battery	Battery (Lithium-Ion)	300 V nominal, 25 Ah, 50 % initial SOC, internal resistance 0.12 Ω , full-charge voltage 349.2 V
Solver	powergui, discrete	Sample time 1 μs , stop time 2 s

Reference operating point. The analysis uses the output plateau read from the model results, $V_{out} = 390 \text{ V}$ and $I_{out} = 20.4 \text{ A}$, as the rated reference point (Section 5). The analytical study treats the battery as a constant-voltage sink at 390 V. The battery block in the model is a 300 V, 25 Ah pack (Section 11 lists this as an alignment item).

4. Resonant design and magnetic coupling

Resonant tuning. A series L-C branch resonates at

$$f_r = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

For the primary, $f_{r,p} = 1 / (2\pi\sqrt{(266.16 \text{ }\mu\text{H} \times 105.74 \text{ nF})}) = \mathbf{30,001 \text{ Hz}}$. For the secondary, $f_{r,s} = 1 / (2\pi\sqrt{(256.79 \text{ }\mu\text{H} \times 109.69 \text{ nF})}) = \mathbf{29,988 \text{ Hz}}$. Both loops are tuned to within 0.04 % of the 30 kHz switching frequency.

Coupling coefficient and coupling reactance.

$$k = \frac{M}{\sqrt{L_p L_s}} \quad (2)$$

$$\omega = 2\pi f_{sw}, \quad \omega M = 2\pi(30,000)(85.46 \times 10^{-6}) \quad (3)$$

$k = 85.46 \text{ }\mu\text{H} / \sqrt{(266.16 \text{ }\mu\text{H} \times 256.79 \text{ }\mu\text{H})} = \mathbf{0.327}$; $\omega = 188,496 \text{ rad/s}$; $\omega M = \mathbf{16.11 \text{ }\Omega}$; $(\omega M)^2 = \mathbf{259.5 \text{ }\Omega^2}$. This is slightly above the 0.2 to 0.3 range typically quoted for EV pads at a 15 cm gap.

Coil reactance and quality factor.

$$Q = \frac{\omega L}{R} \quad (4)$$

$\omega L_p = 50.17 \text{ }\Omega$ and $\omega L_s = 48.40 \text{ }\Omega$. These reactances are the reference for judging how large a coil resistance is physically reasonable (Section 8).

5. Equivalent load and reflected impedance

Output power and DC load.

$$P_{out} = V_{out} I_{out} = 390 \times 20.4 = 7,956 \text{ W} \quad (5)$$

$$R_{dc} = \frac{V_{out}}{I_{out}} = \frac{390}{20.4} = 19.12 \text{ } \Omega \quad (6)$$

A diode bridge followed by a smoothing stage presents an equivalent resistance to the resonant tank that is lower than R_{dc} . For a fundamental-harmonic (FHA) model:

$$R_{ac} = \frac{8}{\pi^2} R_{dc} = 0.8106 \times 19.12 = 15.50 \text{ } \Omega \quad (7)$$

Reflected impedance. With series-series compensation at resonance, the secondary loop reflects a purely resistive impedance into the primary loop:

$$Z_r = \frac{(\omega M)^2}{R_s + R_{ac}} \quad (8)$$

Loop currents. At resonance the secondary current is set by the primary current through the coupling reactance:

$$I_s = \frac{\omega M I_p}{R_s + R_{ac}} \iff I_p = I_s \frac{R_s + R_{ac}}{\omega M} \quad (9)$$

Stage efficiencies. The primary loop delivers a fraction of its power to the reflected resistance, and the secondary loop delivers a fraction to the load:

$$\eta_1 = \frac{Z_r}{R_p + Z_r}, \quad \eta_2 = \frac{R_{ac}}{R_s + R_{ac}}, \quad \eta_{link} = \eta_1 \eta_2 \quad (10)$$

Voltage-source behaviour of the series-series link. When the inverter output voltage V_1 is fixed and the coil resistances are small, the secondary current is approximately

$$I_s \approx \frac{V_1}{\omega M} \quad (11)$$

so a lower mutual inductance (larger gap) raises the secondary current and output power for a fixed inverter voltage. This explains the air-gap trend in Section 9 and why closed-loop power regulation is needed.

6. Efficiency of the near-ideal model (conduction losses)

The as-built model uses 1 m Ω coil resistances and 1 m Ω switch resistances. With $R_s = 1 \text{ m}\Omega$, $Z_r = 259.5 / (0.001 + 15.50) = 16.74 \text{ } \Omega$, and $\eta_1 = \eta_2 = 99.99 \%$, so $\eta_{link} \approx 99.99 \%$. The coil loop is therefore effectively lossless and the system loss is dominated by semiconductor conduction.

Semiconductor conduction losses.

$$P_{diode} = 2(V_f I_{out} + I_{out}^2 R_{on}) = 2(0.8 \times 20.4 + 20.4^2 \times 0.001) = 33.5 \text{ W} \quad (12)$$

$$P_{inv} = 2I_p^2 R_{on} \approx 2(21.9)^2 (0.001) \approx 1.0 \text{ W}, \quad P_{rect} = 2I_{in}^2 R_{on} \approx 1.0 \text{ W} \quad (13)$$

Stage	Devices	Loss expression	Loss (W)
Output diode bridge ($V_f = 0.8 \text{ V}$, $R_{on} = 1 \text{ m}\Omega$)	2 conducting	$2(V_f I_{out} + I_{out}^2 R_{on})$	33.5
HF inverter ($R_{on} = 1 \text{ m}\Omega$)	2 conducting, $I_p \approx 21.9 \text{ A}$	$2 I_p^2 R_{on}$	1.0
Front-end rectifier ($R_{on} = 1 \text{ m}\Omega$, $V_f = 0$)	2 conducting	$2 I_{in}^2 R_{on}$	1.0
Coils (1 mΩ each)	Primary and secondary	$R(I_p^2 + I_s^2)$	1.0
Total			36.5

Efficiency.

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} = \frac{7,956}{7,956 + 35.5} \approx 99.6 \% \quad (14)$$

Including the roughly 1.0 W of coil resistance loss gives 99.54 %. The near-ideal parameters therefore predict an efficiency of about **99.5 to 99.6 %**.

Scope of this figure. This is an analytical conduction-loss estimate for idealised component values. It excludes switching loss, gate-drive loss, diode reverse recovery, core and shield loss, and the frequency-dependent AC resistance of real windings. It should be read as an upper bound, not as a measured or expected system efficiency.

7. Loss budget for the 90 % efficiency target

A practical EV charger loses far more power in its coils and magnetics than the near-ideal model suggests. To represent this, a total loss budget is set from a target efficiency of 90 % (the midpoint of the 88 to 92 % range) and the coil resistance that consumes that budget is solved for.

Step 1. Set the budget.

$$P_{in} = \frac{P_{out}}{\eta} = \frac{7,956}{0.90} = 8,840 \text{ W} \quad (15)$$

$$P_{loss, total} = P_{in} - P_{out} = 8,840 - 7,956 = 884 \text{ W} \quad (16)$$

$$P_{coil} = P_{loss, total} - P_{semi} = 884 - 35.5 \approx 848.5 \text{ W} \quad (17)$$

Loss component	Target allocation (W)
Output diode bridge	33.5
HF inverter conduction	1.0
Front-end rectifier conduction	1.0
Primary and secondary coils	848.5
Total	884.0

Step 2. Fix the currents from the output requirement. The power entering the secondary rectifier includes the 33.5 W diode loss:

$$I_s = \sqrt{\frac{P_{out, link}}{R_{ac}}} = \sqrt{\frac{7,989.5}{15.50}} = 22.70 \text{ A} \quad (18)$$

Step 3. Solve for the equivalent coil resistance. With $R_p = R_s = R$, the primary current from the reflected-impedance relation is $I_p = I_s (R + R_{ac})/\omega M$, and the coil loss is

$$P_{coil} = R(I_p^2 + I_s^2) = R \left[\frac{I_s^2 (R + R_{ac})^2}{(\omega M)^2} + I_s^2 \right] = 848.5 \text{ W} \quad (19)$$

$$R[1.986(R + 15.50)^2 + 515.3] = 848.5 \implies R \approx 0.813 \Omega \quad (20)$$

Step 4. Verify with R = 0.81 Ω .

$$Z_r = \frac{259.5}{0.81 + 15.50} = 15.91 \Omega \quad (21)$$

$$\eta_1 = \frac{15.91}{0.81 + 15.91} = 95.2 \%, \quad \eta_2 = \frac{15.50}{0.81 + 15.50} = 95.0 \% \quad (22)$$

$$\eta_{link} = 95.2 \% \times 95.0 \% = 90.4 \% \quad (23)$$

Quantity	Value
Primary current I_p	22.98 A
Secondary current I_s	22.70 A
Primary coil loss $R I_p^2$	427.7 W
Secondary coil loss $R I_s^2$	417.4 W
Total coil loss	845.1 W
Semiconductor loss	35.5 W
Total loss	880.6 W
Input power $P_{in} = P_{out} + P_{loss}$	8,836.6 W
Overall efficiency	90.0 %

$$\eta_{total} = \frac{7,956}{7,956 + 880.6} = \frac{7,956}{8,836.6} = 90.0 \% \quad (24)$$

Required drive. To deliver 7.956 kW into this lossy link, the inverter fundamental must supply

$$V_1 = I_p (R + Z_r) = 22.98 \times (0.81 + 15.91) \approx 384 \text{ V}_{rms} \quad (25)$$

For a 50 % duty full-bridge square wave, $V_1 = (2\sqrt{2}/\pi) V_{dc} = 0.9 V_{dc}$, which requires $V_{dc} \approx 427 \text{ V}$. With a phase-shift full bridge from a 500 V bus, $V_1 = (4V_{dc}/(\pi\sqrt{2})) \sin(\delta/2)$, and the analytical study finds that a phase shift of $\delta = 116.9^\circ$ delivers the rated power.

8. Interpretation of the equivalent coil resistance

The value $R = 0.81 \Omega$ is an **equivalent** resistance chosen to consume the 848.5 W coil budget in the simplified circuit. It is not the measured winding resistance of a manufactured coil. It lumps together conductor loss (including skin and proximity effect), core and shield loss, and other parasitic losses that are not modelled individually.

A useful physical check is the coil quality factor, $Q = \omega L / R$:

$$Q_p = \frac{\omega L_p}{R} = \frac{50.2}{0.81} \approx 62, \quad Q_s = \frac{\omega L_s}{R} = \frac{48.4}{0.81} \approx 60 \quad (26)$$

Well-built Litz-wire coils with ferrite at tens of kilohertz commonly reach Q between about 100 and 300, which corresponds to R between roughly 0.17 and 0.5 Ω . The 0.81 Ω budget is therefore a deliberately conservative allowance that also covers losses outside the winding. The analytical study repeats the

nominal case with $R = 0.10 \, \Omega$ (low-loss Litz) for comparison: the link efficiency rises from about 90.1 % to about 98.3 % at the same rated power.

Coil resistance per coil	Q (primary)	Link-level efficiency at 7.956 kW	Primary current
1 m Ω (as-built model)	50,000	about 99.99 %	21.9 A
0.10 Ω (low-loss Litz)	502	98.34 %	21.93 A
0.81 Ω (loss-budget equivalent)	62	90.08 %	22.93 A

Efficiencies in the lower two rows come from the fundamental-harmonic model (Section 9) and include diode and switch conduction loss; the first row is the coil-link efficiency only.

9. Analytical robustness study

A fundamental-harmonic (FHA) model of the chain 'DC bus → phase-shift full bridge → compensated coils → diode bridge → constant-voltage battery' was used to examine how the design behaves when operating conditions change. The model is implemented in the repository's Python scripts. The inverter fundamental is

$$V_1 = \frac{4V_{dc}}{\pi\sqrt{2}} \sin\left(\frac{\delta}{2}\right) \quad (27)$$

with $V_{dc} = 500 \, \text{V}$ and phase shift δ . Coupling versus geometry uses a calibrated circular-loop surrogate (radius 31.6 cm), so the air-gap and misalignment values describe trends rather than a manufactured pad. The results below are analytical and are not switching-level Simulink measurements.

9.1 Air-gap sweep (open loop, inverter command fixed at the 15 cm design point, $R = 0.81 \, \Omega$)

Gap (cm)	k	M (μH)	P_{out} (W)	I_{out} (A)	I_1 (A)	Efficiency (%)
10.0	0.452	118.2	5,836	14.96	16.4	92.61
12.5	0.382	99.8	6,862	17.60	19.5	91.39
15.0	0.327	85.5	7,956	20.40	22.9	90.08
17.5	0.283	73.9	9,127	23.40	26.7	88.67
20.0	0.246	64.3	10,382	26.62	30.9	87.16
25.0	0.190	49.6	13,158	33.74	40.8	83.80

With a constant-voltage inverter, larger gaps raise output power and primary current (Equation 11), while efficiency falls. This is the reason a feedback controller is needed.

9.2 Lateral misalignment at 15 cm gap (open loop, $R = 0.81 \, \Omega$)

Lateral (mm)	k	M (μH)	P_{out} (W)	I_1 (A)	Efficiency (%)
0	0.327	85.5	7,956	22.9	90.08
20	0.325	85.0	7,995	23.1	90.03
40	0.320	83.7	8,110	23.4	89.89
60	0.313	81.7	8,301	24.0	89.66
80	0.302	79.0	8,565	24.9	89.35
100	0.290	75.9	8,900	26.0	88.95
120	0.277	72.3	9,308	27.3	88.46

9.3 Frequency sensitivity (open loop, nominal coupling, $R = 0.81 \, \Omega$)

Frequency (kHz)	P _{out} (W)	I ₁ (A)	I ₂ (A)	Efficiency (%)
20.0	0	9.2	0.0	0.00
22.5	0	13.1	0.0	0.00
25.0	0	20.8	0.0	0.00
27.5	10,130	30.2	28.9	87.42
30.0	7,956	22.9	22.7	90.08
32.5	8,162	23.6	23.2	89.83
35.0	10,068	30.4	28.7	87.36
37.5	0	17.0	0.0	0.00
40.0	0	13.1	0.0	0.00

Zero output at 25 kHz and 37.5 kHz means the rectifier does not conduct because the available voltage is below the battery voltage. Power peaks near 27.5 kHz and 35 kHz (frequency splitting) and the efficiency maximum is 90.20 % at 31.0 kHz, which supports operation at the 30 kHz design point.

9.4 Compensation topology comparison (R = 0.81 Ω, same coils, same battery)

Topology	C _p (nF)	L _f (μH)	V ₁ (V)	P _{out} (W)	Efficiency (%)	Coil current (A)
SS	105.7	0.0	383.6	7,956	90.08	22.9
SP	118.2	0.0	178.3	7,956	64.93	68.5
PS	174.6	305.5	450.2	7,956	90.08	22.9
PP	164.1	595.4	450.2	7,956	64.97	68.5

Series-series (SS) reaches the rated point with the lowest coil current and about 90 % efficiency. Parallel-secondary arrangements need about 68 A in the transmitter coil at this coupling, which gives roughly 65 % efficiency. As a check, for a parallel secondary $I_1 = V_{\text{rect}} L_s / (M R_{\text{eq}}) = 351 \times 256.79 / (85.46 \times 15.49) \approx 68 \text{ A}$.

9.5 Closed-loop power regulation

A PI controller adjusts the inverter phase shift δ to hold the measured output power at the 7.956 kW reference. For the air-gap sequence 15, 20, 12.5, 15 cm, open-loop power varies from about 6.86 to 10.38 kW, while the regulated output returns to the reference after each step (settling inside a $\pm 2 \%$ band in about 9, 16 and 12 ms in the quasi-static model). For lateral displacement 0 to 80 mm and back, the regulated output stays within about $\pm 0.1 \%$ of the reference. Regulation restores power but does not remove the efficiency penalty of weak coupling, because the coil currents still rise. Power regulation and efficiency optimisation are different objectives.

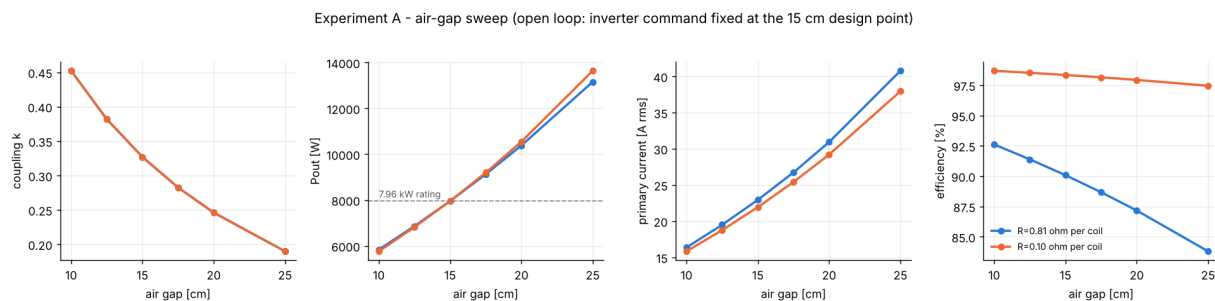


Figure 4. Experiment A: air-gap sweep, open loop.

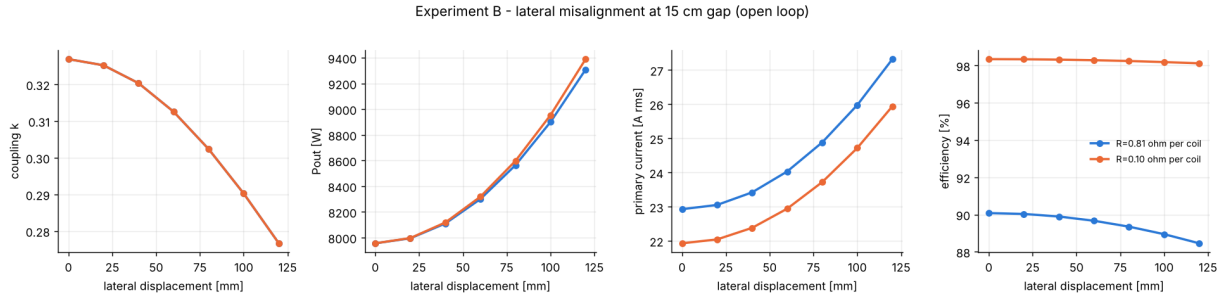


Figure 5. Experiment B: lateral misalignment at 15 cm gap.

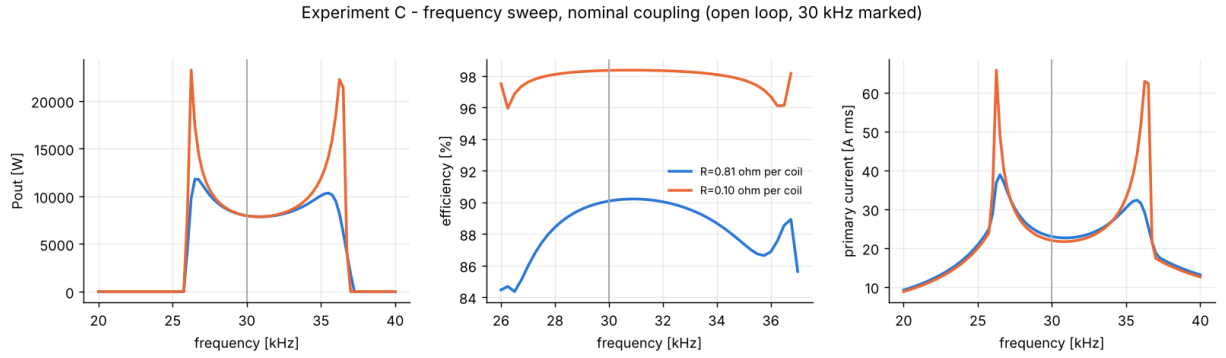


Figure 6. Experiment C: frequency sweep at nominal coupling.

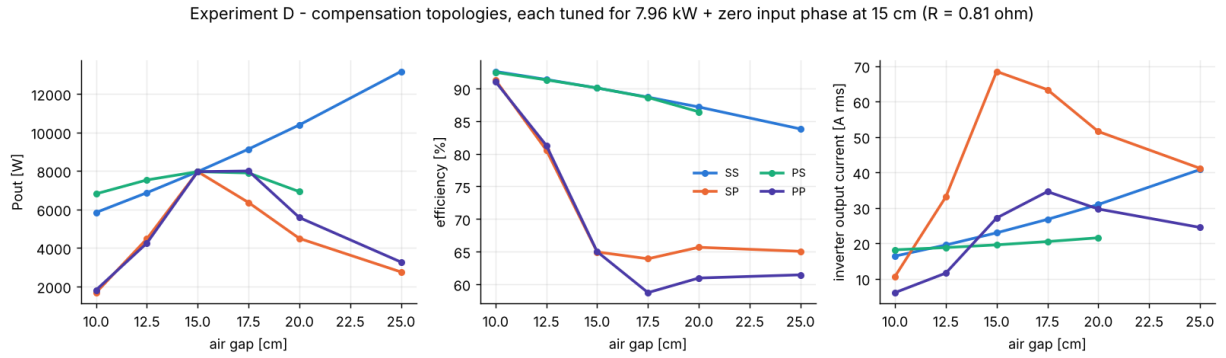


Figure 7. Experiment D: compensation-topology comparison.

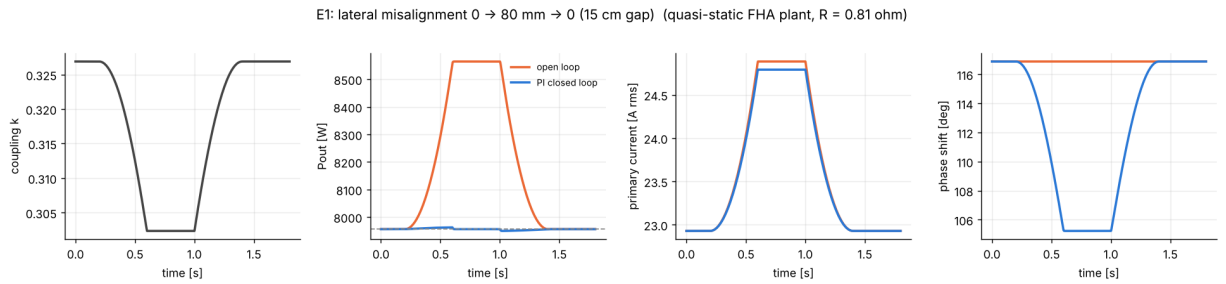


Figure 8. Experiment E1: closed-loop regulation under lateral misalignment.

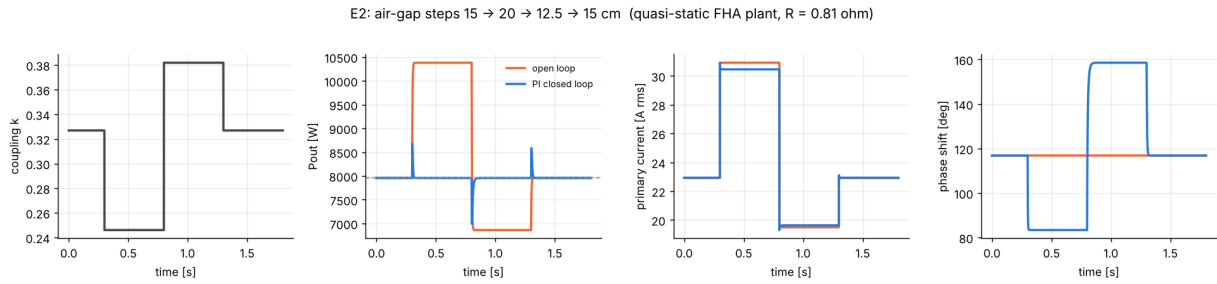


Figure 9. Experiment E2: closed-loop regulation under air-gap steps.

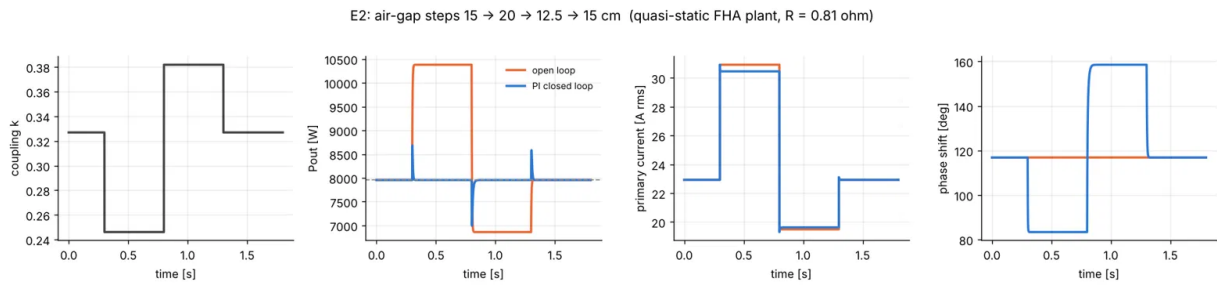


Figure 10. Air-gap control figure (user-provided).

10. Captured Simulink waveforms

The scope captures below document the switching-level model at each stage of the power-transfer chain. Values quoted in the captions are the statistics displayed in the scope windows.

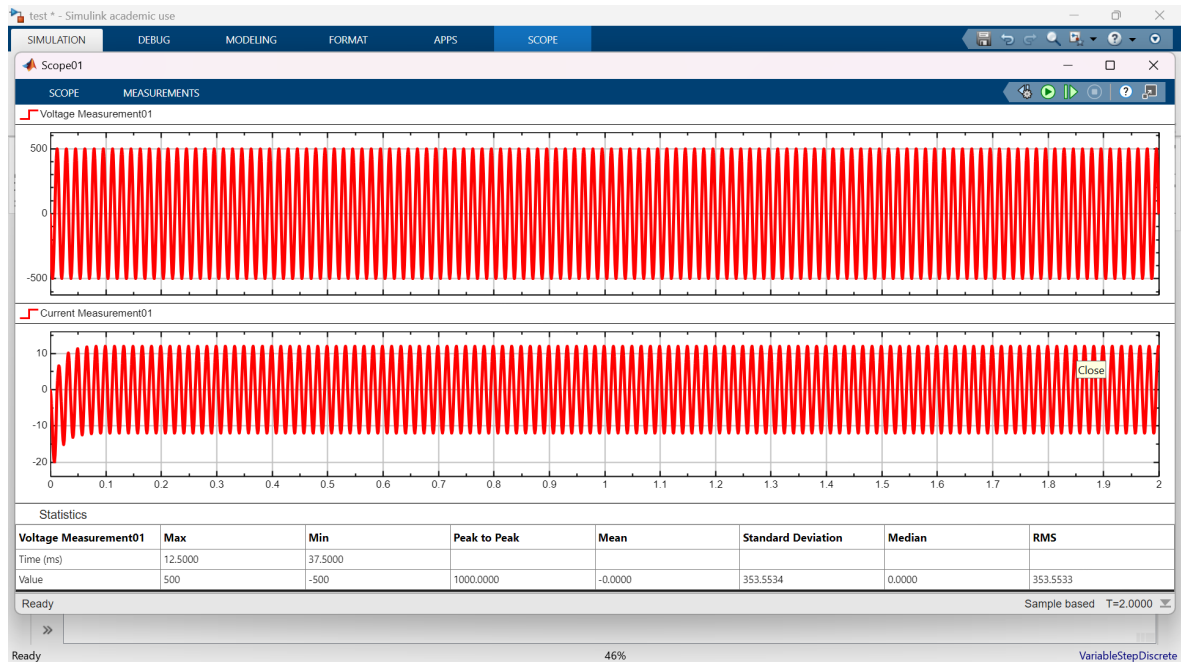


Figure 11. Scope01: source voltage (500 V peak, 353.55 V rms, 60 Hz) and source current.

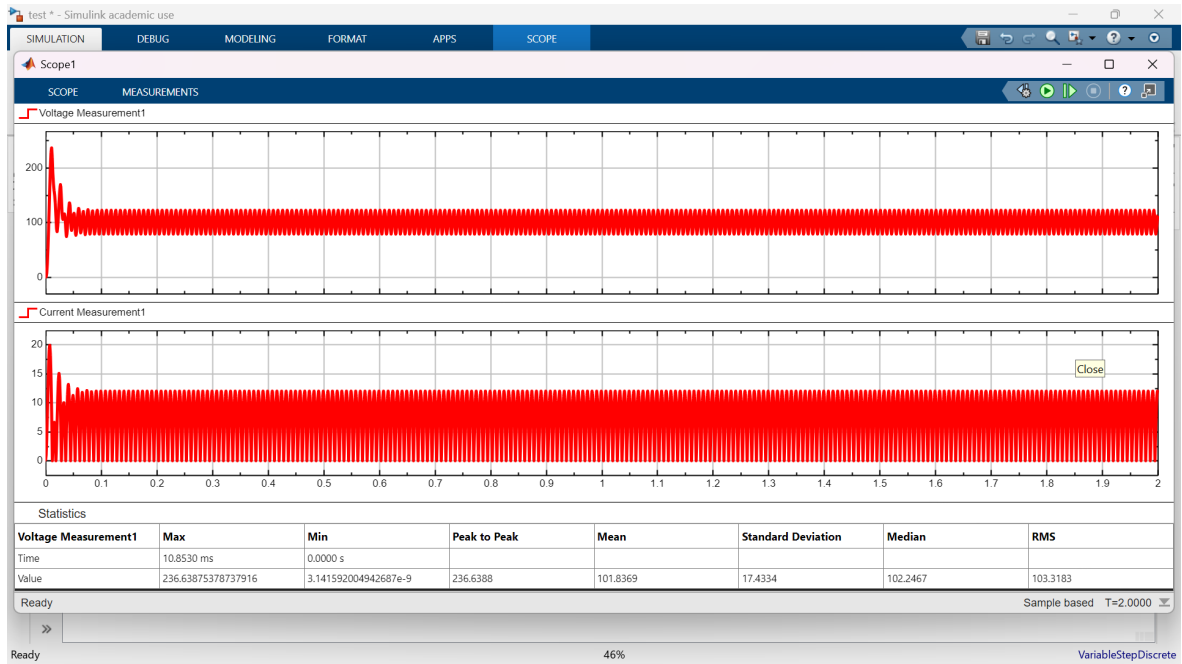


Figure 12. Scope1: primary-side voltage and current.

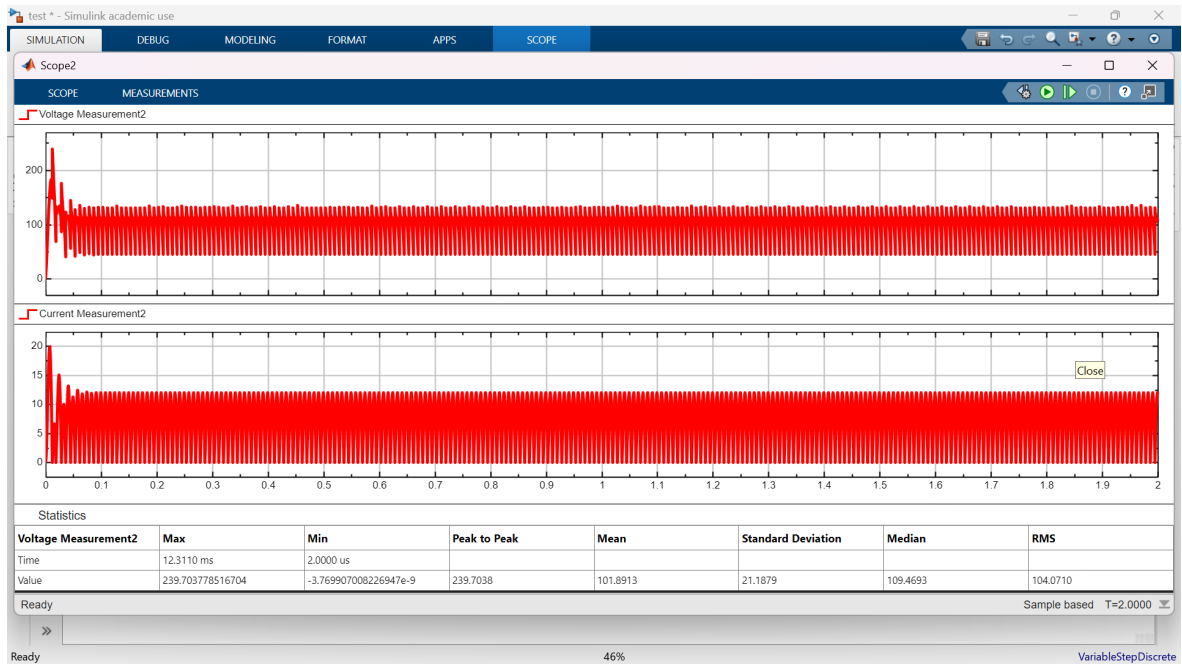


Figure 13. Scope2: compensation-network voltage and current.

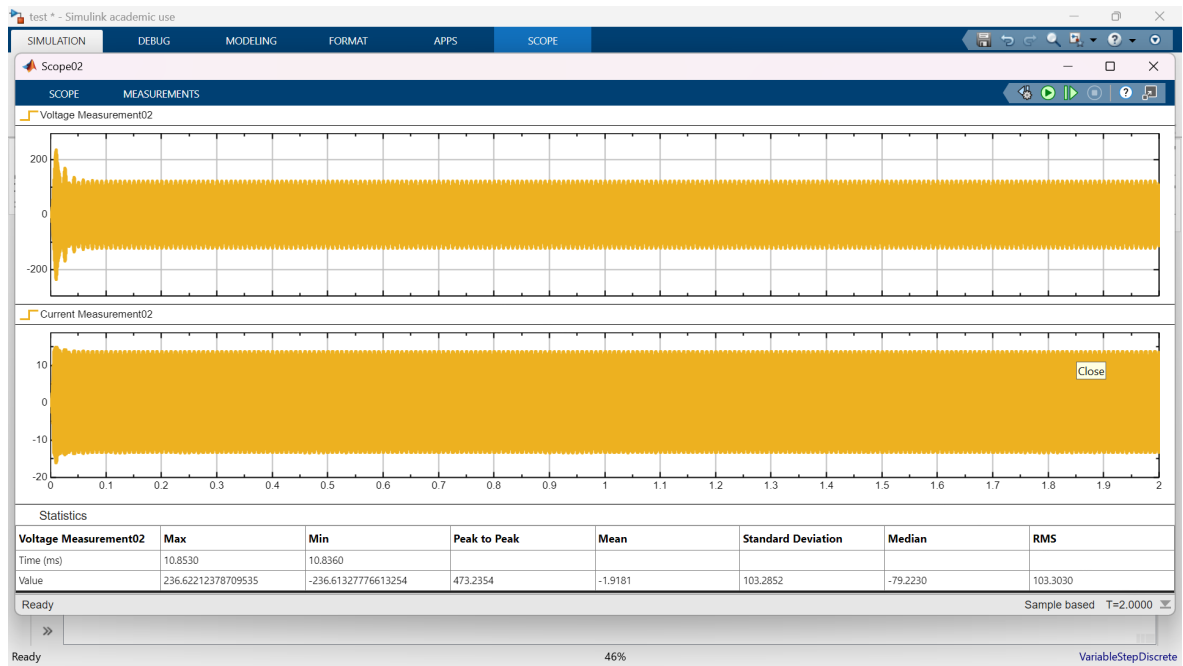


Figure 14. Scope02: secondary-side voltage and current.

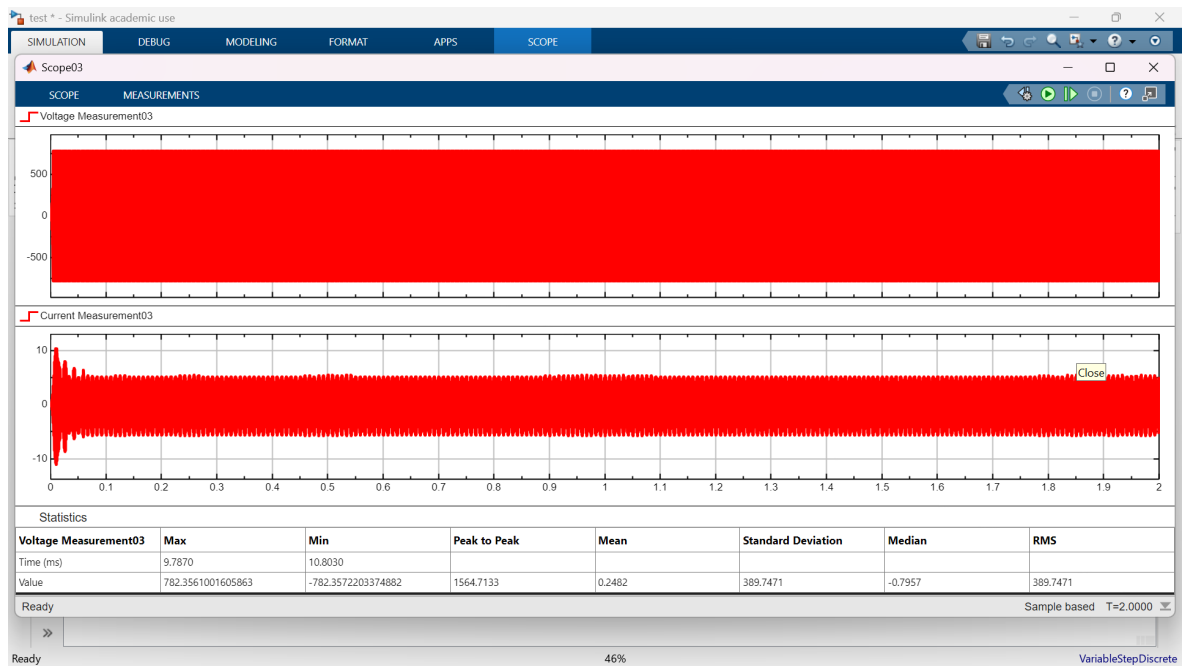


Figure 15. Scope03: secondary resonant network at the rectifier input (voltage peaks of about 782 V, 389.7 V rms).

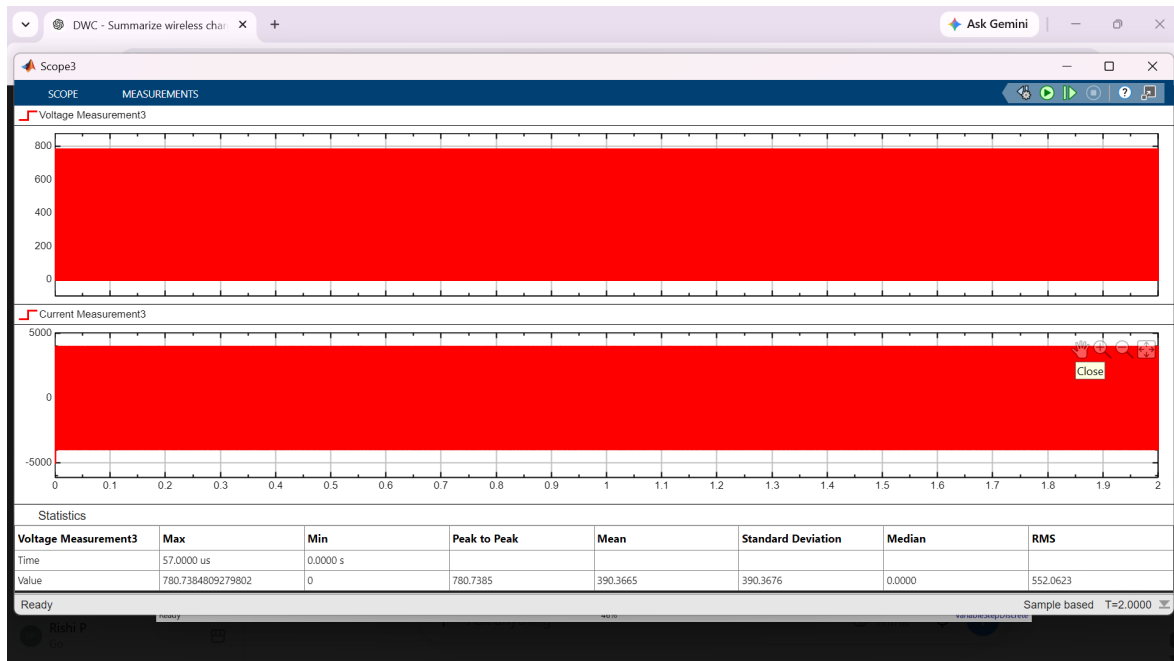


Figure 16. Scope3: output voltage and current. The voltage statistics show a mean of 390.37 V with a maximum of 780.7 V.

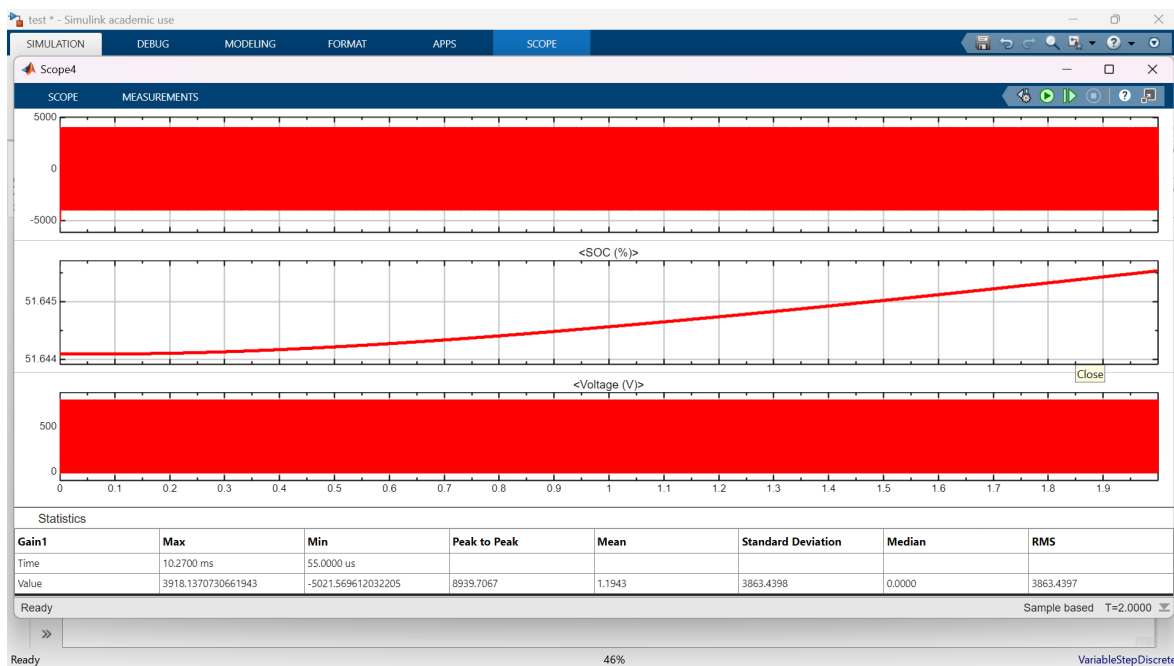


Figure 17. Scope4: battery current, state of charge (rising over the run) and battery voltage.

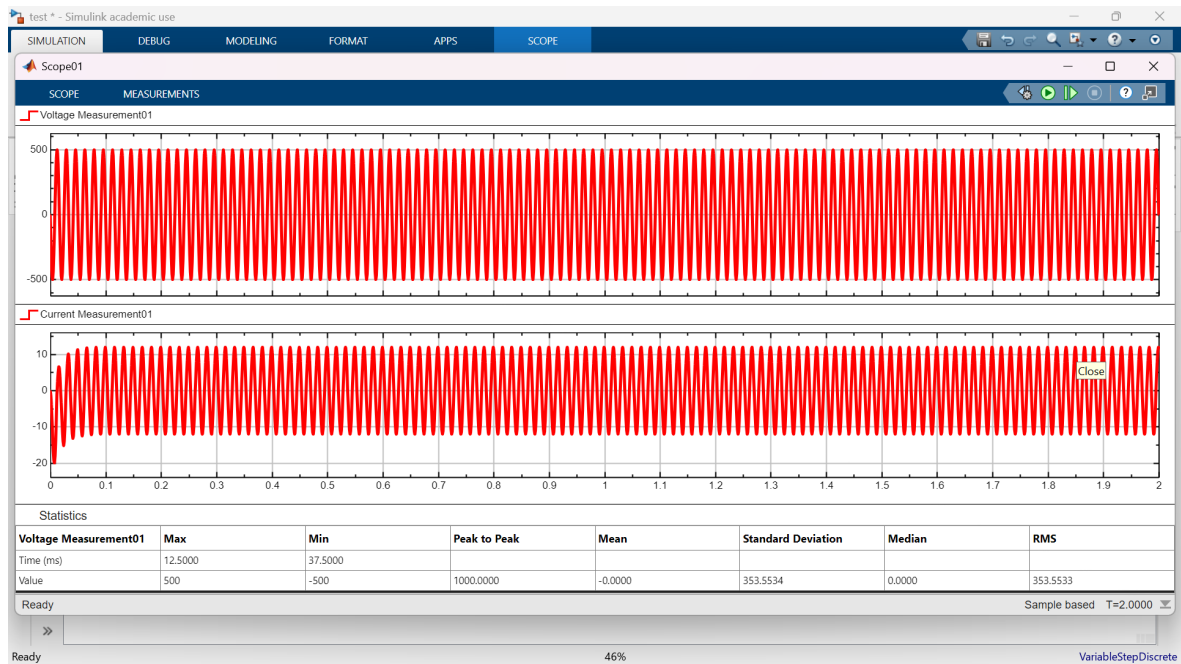


Figure 18. Combined output waveforms of the model.

11. Verification status and recommended next steps

The table separates what is established by calculation from what remains to be confirmed by averaged power measurement in the switching-level model.

Result	Basis	Status
Resonant frequencies 30,000 Hz and 29,990 Hz	Exact model values, Equation 1	Calculated
Coupling coefficient $k = 0.327$; $\omega M = 16.11 \Omega$	Model M , L_p , L_s	Calculated
Near-ideal efficiency 99.5 to 99.6 %	Conduction-loss budget, 1 m Ω coils	Analytical estimate
90 % efficiency with $R = 0.81 \Omega$ per coil	Loss budget and reflected-impedance model	Design target, analytically consistent
Air gap, misalignment, frequency, topology, PI control	FHA Python model	Analytical (not switching-level)
Average input and output power in the Simulink model	Averaged $v \cdot i$ over a steady-state window	To be measured and added

Recommended next steps.

- Average-power measurement.** Add Product and Mean blocks ($v \times i$, averaged over an integer number of line and switching periods) at the source and at the battery terminals, and compute $\eta = P_{\text{battery}} / P_{\text{source}}$.
- Battery and DC bus alignment.** The analysis assumes a 390 V sink and a bus able to supply V_{dc} of about 426 V (50 % duty) or a phase-shifted bridge on 500 V. The battery block is rated 300 V nominal with 349.2 V full-charge voltage, and the 300 μF capacitor is small for a single-phase 8 kW front end. Matching the battery voltage and capacity, and sizing the DC-link capacitance for low ripple, makes the model consistent with the 7.956 kW point.
- Output filtering.** Place an LC filter between the output rectifier and the battery so that the measured output current and voltage are smooth DC quantities.
- Insert coil resistance.** Replace the 1 m Ω coil resistances by 0.81 Ω (budget case) and by 0.1 to 0.3 Ω (Litz-realistic case), and compare the measured power and efficiency with Section 9.

5. **Dynamic charging.** For a vehicle travelling over segmented transmitters (Figure 2b), sweep $k(x)$ along the track. With series-series compensation and a constant-voltage inverter, a segment with no vehicle has $M \rightarrow 0$ and a primary current approaching V_1/R_p (about 474 A at $R = 0.81 \Omega$), so each segment must be switched off until the vehicle is detected.

6. **Real coil geometry.** Replace the circular-loop surrogate with a pad design and add thermal and magnetic-field limits.

12. Repository files referenced

File	Content
simulation/Resonant_Power_Transfer.slx	Switching-level Simulink / Simscape Electrical model
design/resonance_calculations.m	Resonance, coupling and 90 % loss-budget script
analysis/wireless_power_transfer_model.py	FHA model, coupling geometry
analysis/compensation_topologies.py	SS / SP / PS / PP solver
analysis/control_tuning.py	PI tuning for phase-shift regulation
analysis/run_experiments.py, run_log.txt, results/	Parametric studies, log and CSV results
figures/simulink, figures/waveforms, figures/robustness	Model views, scope captures, analysis plots

Appendix A. Simulink model views

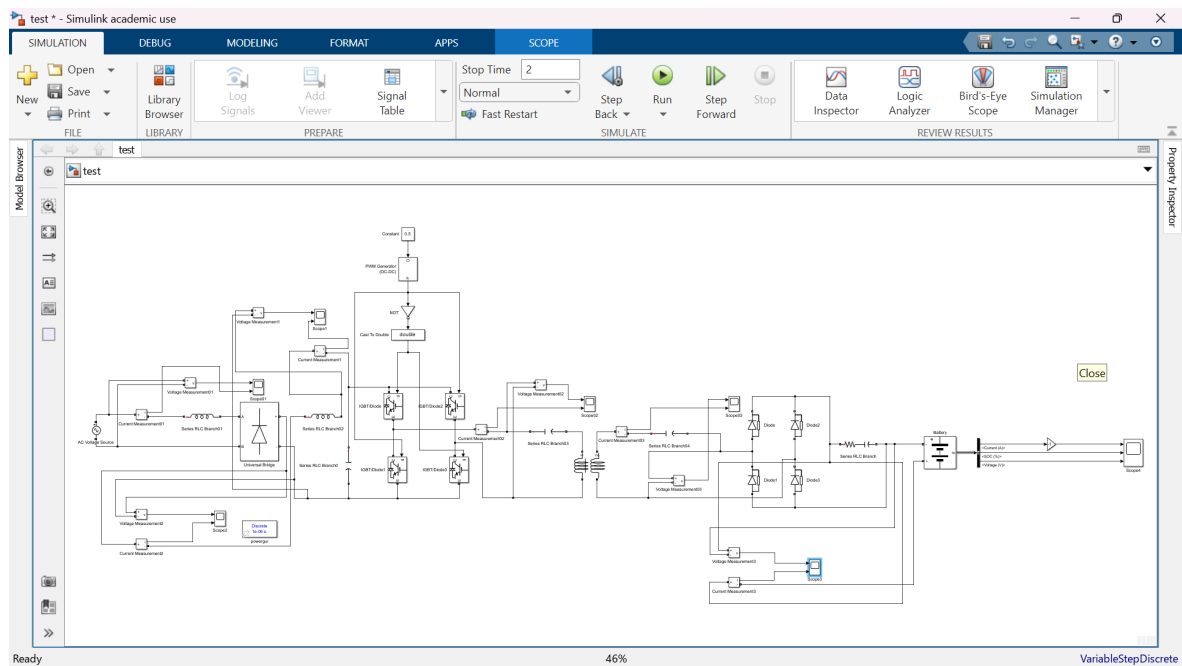


Figure 19. Full Simulink model of the wireless charging system.

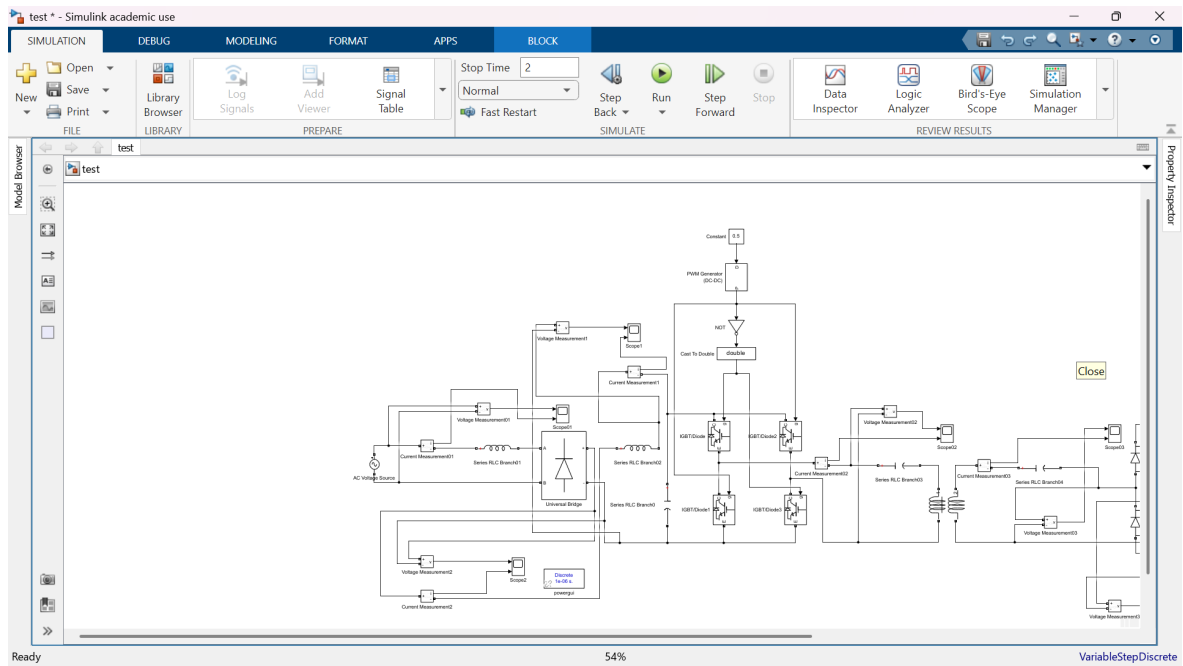


Figure 20. Inverter and primary resonant stage.

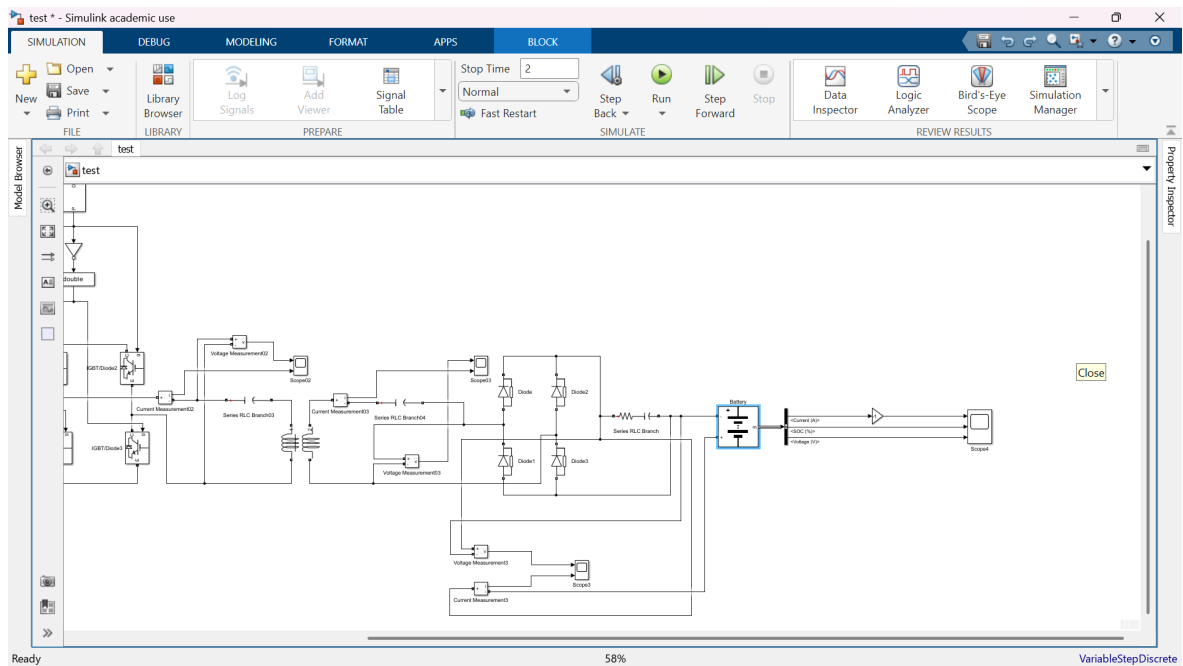


Figure 21. Secondary resonant network, output rectifier and battery stage.